Algebra Homework 2

Problem 1. Compute the integral closure of $\mathbb{Z}$

(i) in $\mathbb{Q}(\sqrt{D})$, where D is a squarefree integer,

(ii) in the normal hull of $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$.

Proof. (i) If $D = 1$, the answer is $\mathbb{Z}$. Suppose $D \neq 1$ and write $x = a + b\sqrt{D}$, with $a, b \in \mathbb{Q}$. By the trace and norm criterion, x is integral if and only if

$$2a \in \mathbb{Z}, \qquad a^2 - Db^2 \in \mathbb{Z}.$$

These conditions imply

$$D(2b)^2 = (2a)^2 - 4(a^2 - Db^2) \in \mathbb{Z}.$$

The square of the reduced denominator of $2b$ divides D, so squarefreeness gives $2b \in \mathbb{Z}$. The remaining condition is

$$(2a)^2 - D(2b)^2 \equiv 0 \pmod 4.$$

If $D \equiv 2, 3 \pmod 4$, both $2a$ and $2b$ must be even. If $D \equiv 1 \pmod 4$, they must have the same parity, and then

$$x = (a - b) + (2b)\frac{1 + \sqrt{D}}{2}.$$

Conversely, $\sqrt{D}$ is integral, and $(1 + \sqrt{D})/2$ satisfies $T^2 - T + (1 - D)/4 = 0$ when $D \equiv 1 \pmod 4$. Thus

$$\mathcal{O}_{\mathbb{Q}(\sqrt{D})} = \begin{cases} \mathbb{Z}\left[\dfrac{1 + \sqrt{D}}{2}\right], & D \equiv 1 \pmod 4, \\ \mathbb{Z}[\sqrt{D}], & D \equiv 2, 3 \pmod 4. \end{cases}$$

(ii) Let $\alpha = \sqrt[3]{2}$ and $\omega = (-1 + \sqrt{-3})/2$. The normal hull is $K = \mathbb{Q}(\alpha, \omega)$, since the roots of $T^3 - 2$ are $\alpha, \omega\alpha, \omega^2\alpha$. Eisenstein at 2 gives $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, and $\omega \notin \mathbb{Q}(\alpha) \subseteq \mathbb{R}$, so $[K : \mathbb{Q}] = 6$.

We first compute the integers of $E = \mathbb{Q}(\alpha)$. For $y = a + b\alpha + c\alpha^2$, with $a, b, c \in \mathbb{Q}$, the characteristic polynomial of multiplication by y is

$$T^3 - 3aT^2 + (3a^2 - 6bc)T - (a^3 + 2b^3 + 4c^3 - 6abc).$$

If y is integral, these coefficients are integers. Taking the traces of $y, \alpha y, \alpha^2 y$ gives $3a, 6c, 6b \in \mathbb{Z}$, hence $6y \in \mathbb{Z}[\alpha]$.

Suppose $(a + b\alpha + c\alpha^2)/2$ is integral, with $a, b, c \in \mathbb{Z}$. Its trace forces a to be even, and its norm gives

$$a^3 + 2b^3 + 4c^3 - 6abc \equiv 0 \pmod 8.$$

Reduction modulo 4 forces b to be even, and reduction modulo 8 then forces c to be even. If $(a + b\alpha + c\alpha^2)/3$ is integral, its second characteristic coefficient and norm give

$$a^2 - 2bc \equiv 0 \pmod 3, \qquad a^3 + 2b^3 + 4c^3 - 6abc \equiv 0 \pmod{27}.$$

Modulo 3, the second congruence says $a - b + c = 0$. Substituting $b \equiv a + c$ into the first gives $(a - c)^2 \equiv 0$, so $a \equiv c$ and $b \equiv -c \pmod 3$. Writing $a = c + 3h$ and $b = -c + 3k$, the norm numerator is $9c^3$ modulo 27. Thus $3 \mid c$, and also $3 \mid a, b$. Applying the two denominator calculations to $3y$ and $2y$ shows that both lie in $\mathbb{Z}[\alpha]$, hence so does y. Therefore

$$A := \mathcal{O}_E = \mathbb{Z}[\alpha].$$

Write $x = a + b\omega$, with $a, b \in E$. Since

$$\mathrm{Tr}_{K/E}(x) = 2a - b, \qquad \mathrm{N}_{K/E}(x) = a^2 - ab + b^2,$$

x is integral if and only if both expressions belong to A. For integral x, the trace of ωx also gives $-a - b \in A$. Hence $a + b, 3b \in A$, and we can write

$$x = u + \frac{v(\omega - 1)}{3}, \qquad u, v \in A.$$

Its trace is $2u - v$ and its norm is $u^2 - uv + v^2/3$. Thus

$$\mathcal{O}_K = \left\{ u + \frac{v(\omega - 1)}{3} : u, v \in A, \quad v^2 \in 3A \right\}.$$

Now

$$A/3A \simeq \mathbb{F}_3[U]/(U^3), \qquad \alpha + 1 \longmapsto U.$$

In this quotient, $(r + tU + wU^2)^2 = 0$ forces $r = t = 0$. Consequently, $v^2 \in 3A$ if and only if $v \in (\alpha + 1)^2 A + 3A$. Since

$$(\alpha + 1)^3 = 3(\alpha^2 + \alpha + 1), \qquad (\alpha - 1)(\alpha^2 + \alpha + 1) = 1,$$

we have $3A = (\alpha + 1)^3 A$. Therefore

$$\mathcal{O}_K = A \oplus A\eta, \qquad \eta = \frac{\omega - 1}{\alpha + 1}.$$

Here we used $(\alpha + 1)^2(\omega - 1)/3 = (\alpha^2 + \alpha + 1)\eta$, whose coefficient is a unit of A. An integral basis is

$$1,\ \alpha,\ \alpha^2,\ \eta,\ \alpha\eta,\ \alpha^2\eta.$$

To express the answer over $F = \mathbb{Q}(\omega)$, put

$$R = \mathcal{O}_F = \mathbb{Z}[\omega], \qquad \pi = 1 - \omega, \qquad s = 1 + 2\omega = \sqrt{-3} = \omega\pi,$$

and define

$$\beta = \frac{(\alpha + 1)^2}{\pi}, \qquad \gamma = \frac{s}{\alpha + 1}.$$

The identities

$$\begin{aligned} \omega &= 1 + (\alpha + 1)\eta, \\ \alpha\omega &= \alpha + (\alpha + \alpha^2)\eta, \\ \beta &= (\alpha + 1)^2 + (1 + \alpha + \alpha^2)\eta \end{aligned}$$

give the integral basis $1, \alpha, \alpha^2, \omega, \alpha\omega, \beta$, since the change of basis has determinant 1.

Moreover,

$$\omega\beta = \beta - \alpha^2 - 2\alpha - 1,$$

so this basis and $1, \omega, \alpha, \alpha\omega, \beta, \omega\beta$ span the same $\mathbb{Z}$-module. Hence

$$\mathcal{O}_K = R \oplus R\alpha \oplus R\beta.$$

Finally,

$$\gamma^2 = 1 - \alpha^2, \qquad \alpha = 2 + s\gamma - \gamma^2, \qquad \beta = -\omega(\gamma + s\alpha).$$

Since $\gamma = -\omega^2\beta - s\alpha \in \mathcal{O}_K$ and $\alpha, \beta \in R[\gamma]$, we obtain $\mathcal{O}_K = R[\gamma]$.

A local proof uses the following Eisenstein lemma.

Lemma. *Let D be a DVR and $f \in D[T]$ a monic Eisenstein polynomial. If θ is a root, then $D[\theta]$ is a DVR with maximal ideal (θ).*

Let ϖ be a uniformizer and $n = \deg f$. Eisenstein makes $D[\theta]$ a domain, finite over D. Every maximal ideal contracts to (ϖ), and

$$D[\theta]/\varpi D[\theta] \simeq (D/\varpi D)[T]/(T^n)$$

has a unique maximal ideal. Hence $D[\theta]$ is Noetherian and local, with maximal ideal (ϖ, θ). The equation $f(\theta) = 0$ puts its constant coefficient, a unit times ϖ, in (θ). Thus the maximal ideal is (θ), and $D[\theta]$ is a DVR.

In particular, if f over a UFD is Eisenstein at a prime element ϖ, the power-basis coefficients of an integral element lie in the localization at (ϖ): this follows by applying the lemma there, since a DVR is integrally closed. For E, the polynomials of α and $\alpha + 1$ are

$$T^3 - 2, \qquad T^3 - 3T^2 + 3T - 3,$$

Eisenstein at 2 and 3, respectively. Together with $6\mathcal{O}_E \subseteq \mathbb{Z}[\alpha]$, the lemma gives $\mathcal{O}_E = \mathbb{Z}[\alpha]$, replacing the congruence calculations above. The bases $1, \alpha, \alpha^2$ and $1, \alpha + 1, (\alpha + 1)^2$ differ by a unimodular integral matrix.

For a proof of $\mathcal{O}_K = R[\gamma]$ independent of the computation over E, recall that R is Euclidean for the norm $m^2 - mn + n^2$, and

$$R/(2) \simeq \mathbb{F}_4, \qquad R/(s) \simeq \mathbb{F}_3, \qquad s^2 = -3.$$

If $x = a + b\alpha + c\alpha^2 \in \mathcal{O}_K$, with $a, b, c \in F$, relative traces give $3a, 6b, 6c \in R$. Eisenstein at 2 excludes 2 from their denominators, so $a, b, c \in R[1/s]$. The equation

$$\gamma^3 - s\gamma^2 - 3\gamma + s = 0$$

is Eisenstein at s, and $F(\gamma) = K$ because $\alpha = 2 + s\gamma - \gamma^2$. Rewriting x gives

$$x = (a + 2b + c) + sb\gamma - (b + c)\gamma^2.$$

The coefficients lie in $R[1/s]$; the lemma also puts them in $R_{(s)}$. Since $R[1/s] \cap R_{(s)} = R$, they belong to R. This proves $\mathcal{O}_K = R[\gamma]$.

One can use β at the prime above 3 instead. Its equation

$$\beta^3 + \omega^2\pi\beta^2 + 3\omega^2\beta - \omega\pi = 0$$

is Eisenstein at π, since $\pi^2 = -3\omega$. Moreover,

$$2\alpha = 1 + 2\pi\beta + \omega\beta^2.$$

Since 2 is a unit in $R_{(\pi)}$, we obtain

$$R_{(\pi)}[\beta] = R_{(\pi)} \oplus R_{(\pi)}\alpha \oplus R_{(\pi)}\beta.$$

At (2), the definition of β gives $R_{(2)} + R_{(2)}\alpha + R_{(2)}\beta = R_{(2)}[\alpha]$. At every other prime, $R[\alpha]$ is already maximal by the trace bound. Thus $R \oplus R\alpha \oplus R\beta$ is maximal at every prime, giving the same answer.

The discriminant provides another way to restrict the relevant primes to those above 2 and 3. Set

$$\theta = \frac{1 - \alpha + \alpha^2}{s} = -\gamma.$$

The identity $\alpha = 2 + s\gamma - \gamma^2$ shows that $R + R\alpha + R\theta = R[\gamma]$. Relative to $1, \alpha, \alpha^2$, the basis change has determinant $1/s$, so

$$\operatorname{disc}_{K/F}(1, \alpha, \theta) = \frac{-108}{s^2} = 36.$$

Only primes dividing this discriminant can divide the index; the preceding Eisenstein arguments establish maximality there. The tower formula then gives

$$\operatorname{disc}(K) = (-3)^3 \operatorname{N}_{F/\mathbb{Q}}(36) = -2^4\, 3^7.$$

There is also a global denominator argument. Put $S = R \oplus R\alpha \oplus R\beta = R[\gamma]$. The polynomial of γ and the displayed change-of-basis identities show that S is a ring of algebraic integers without assuming maximality. For nonzero $z \in S$, the determinant of multiplication by z on this $\mathbb{Z}$-lattice gives

$$|S/zS| = |\operatorname{N}_{K/\mathbb{Q}}(z)|.$$

Now $\operatorname{N}_{K/\mathbb{Q}}(\alpha) = 4$ and $\operatorname{N}_{K/\mathbb{Q}}(\beta) = 3$, since $\operatorname{N}_{E/\mathbb{Q}}(\alpha + 1) = \operatorname{N}_{F/\mathbb{Q}}(\pi) = 3$. Thus $S/\beta S \simeq \mathbb{F}_3$. The unital map $R/(2) \to S/\alpha S$ is injective because its source is $\mathbb{F}_4$, and it is surjective because its target also has four elements. Hence α and β are prime elements of S.

The identities

$$2 = \alpha^3, \qquad 3 = -(\alpha - 1)^4\beta^6$$

and $6\mathcal{O}_K \subseteq S$ allow every $x \in \mathcal{O}_K$ to be written as $y/(\alpha^m\beta^n)$ with $y \in S$ and $m, n \geq 0$. Choose such an expression with $m + n$ minimal. Clearing denominators in a monic equation of degree d for x gives $y^d \in \alpha^m\beta^n S$. If $m > 0$, primality forces $\alpha \mid y$; if $n > 0$, it forces $\beta \mid y$. Either contradicts minimality, so $m = n = 0$ and $x \in S$.

Collecting the equivalent descriptions,

$$\begin{aligned}\mathcal{O}_K &= \mathbb{Z}[\alpha] \oplus \mathbb{Z}[\alpha]\eta \\ &= \mathbb{Z}[\omega] \oplus \mathbb{Z}[\omega]\alpha \oplus \mathbb{Z}[\omega]\beta \\ &= \mathbb{Z}[\omega, \gamma] = \mathbb{Z}[\omega, \theta] = \mathbb{Z}[\alpha, \omega, \beta].\end{aligned}$$

The corresponding integral bases include

$$\begin{aligned}&1,\ \alpha,\ \alpha^2,\ \eta,\ \alpha\eta,\ \alpha^2\eta, \\ &1,\ \alpha,\ \alpha^2,\ \omega,\ \alpha\omega,\ \beta, \\ &1,\ \omega,\ \gamma,\ \omega\gamma,\ \gamma^2,\ \omega\gamma^2.\end{aligned}$$

Problem 2. Let $\mathcal{O}$ be the integral closure of $\mathbb{Z}$ in $\mathbb{C}$, let p be a prime integer, and let $\mathfrak{P}$ be a maximal ideal of $\mathcal{O}$ containing p. Prove that $\mathcal{O}/\mathfrak{P}$ is an algebraic closure of $\mathbb{F}_p$.

Proof. The field $\mathcal{O}/\mathfrak{P}$ has characteristic p. Reducing a monic equation over $\mathbb{Z}$ for each element of $\mathcal{O}$ shows that $\mathcal{O}/\mathfrak{P}$ is algebraic over $\mathbb{F}_p$.

Let $f \in (\mathcal{O}/\mathfrak{P})[T]$ be monic, and lift its coefficients to obtain a monic polynomial $F \in \mathcal{O}[T]$. Every complex root of F is integral over $\mathcal{O}$, hence over $\mathbb{Z}$, and therefore belongs to $\mathcal{O}$. Reducing the factorization of F into linear factors modulo $\mathfrak{P}$ gives a factorization of f into linear factors. Thus $\mathcal{O}/\mathfrak{P}$ is algebraically closed and is an algebraic closure of $\mathbb{F}_p$.

Problem 3. Let $I = ((X-1)^2)$ in $\mathbb{Z}[X]$ and $J = ((X-1)^2)$ in $\mathbb{Q}[X]$. Set

$$A = \mathbb{Z}[X]/I, \qquad B = \mathbb{Q}[X]/J.$$

(i) Prove that $J \cap \mathbb{Z}[X] = I$.

(ii) Find the integral closure of A in B.

Proof. (i) Divide $f \in J \cap \mathbb{Z}[X]$ by the monic polynomial $(X-1)^2$ in $\mathbb{Z}[X]$. The remainder is also the remainder in $\mathbb{Q}[X]$, so it is zero. Hence $J \cap \mathbb{Z}[X] = I$.

(ii) Let $\varepsilon = (X-1)$, we have

$$A = \mathbb{Z} \oplus \mathbb{Z}\varepsilon \subseteq B = \mathbb{Q} \oplus \mathbb{Q}\varepsilon, \qquad \varepsilon^2 = 0.$$

If $r + s\varepsilon \in B$ is integral over A, applying $B \to \mathbb{Q}$, $r + s\varepsilon \mapsto r$, to a monic equation shows that r is integral over $\mathbb{Z}$, hence $r \in \mathbb{Z}$.

Conversely, for $r \in \mathbb{Z}$ and $s \in \mathbb{Q}$, the element $r + s\varepsilon$ satisfies $(T-r)^2 \in A[T]$. Therefore the integral closure is $\mathbb{Z} \oplus \mathbb{Q}\varepsilon$.